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Prof. Rodney Van Meter, Editor-in-Chief
IEEE Transactions on Quantum Engineering (IEEE TQE)
IEEE Quantum Initiative • IEEE Computer Society • IEEE Photonics Society

SUBJECT: Submission of Original Research Manuscript — "Orbit-Tied Quantized Neural Belief Propagation for Real-Time Sliding-Window Decoding of Bivariate-Bicycle Quantum Codes"

Dear Prof. Van Meter and the Editorial Board of *IEEE Transactions on Quantum Engineering*,

I am pleased to submit our original research manuscript, titled **"Orbit-Tied Quantized Neural Belief Propagation for Real-Time Sliding-Window Decoding of Bivariate-Bicycle Quantum Codes,"** for peer-reviewed publication as a Regular Paper in the *IEEE Transactions on Quantum Engineering*. The manuscript consists of 12 pages in standard IEEE Transactions two-column format, comprising 7 multi-panel publication figures, 5 comparative benchmark tables, formal mathematical proofs, synthesizeable hardware designs, and 59 comprehensive citations.

Background and Motivation: Quantum Low-Density Parity-Check (QLDPC) bivariate-bicycle (BB) codes, pioneered by Bravyi et al. (2024), represent a transformative milestone toward scalable fault-tolerant quantum computation, enabling an order-of-magnitude reduction in physical qubit overhead compared to standard 2D topological surface codes. However, translating BB codes into operational quantum processors faces a critical decoding bottleneck: their dense, non-local Tanner graphs produce complex 3D space-time Detector Error Models (DEM) with thousands of correlated fault mechanisms. Conventional decoders (such as BP+OSD) require hundreds of microseconds on high-end CPUs, causing catastrophic backlog and buffer overflow under the strict ~1 μs coherence budget of superconducting quantum processors.

Key Scientific & Engineering Contributions:

1. **Torus Automorphism Equivariance (Theorem 2):** We prove that the 2D torus translation group $G = \mathbb{Z}_\ell imes \mathbb{Z}_m$ forms an exact automorphism group of the space-time DEM hypergraph. By tying neural BP weights across automorphism orbits, we collapse 7,308 DEM fault mechanisms into 203 orbits ($36.0 imes$ reduction) on $[[72,12,6]]$ and 14,616 mechanisms into 203 orbits ($72.0 imes$ reduction) on $[[144,12,12]]$, guaranteeing mathematical equivariance and eliminating untied neural overfitting.

2. **Multiplier-Free 6-Bit Dyadic CSD Architecture (Theorem 3):** To eliminate costly Digital Signal Processing (DSP) multiplier blocks entirely ($ ext{DSP}=0$), we prove dyadic fixed-point bounds ($[-7.75, +7.75]$, $\Delta=0.25$) and implement all variable- and check-node updates using two-term Canonical Signed-Digit (CSD) shift-and-add arithmetic, enabling ultra-compact FPGA deployment.

3. **Multi-Stage Margin Gating & Streaming Sliding Window:** We engineer a confidence-margin fast-path gate (G_3) within a continuous sliding window ($W=3, F=1$). Gate G_3 routes 75.6% of syndrome windows with zero logical errors ($P_{L, ext{accept}} = 0.0000$), reducing queue backlog to $6.0 imes 10^{-5}$ and delivering a pipeline logical error rate of $P_{ ext{pipe}} = 1.90 imes 10^{-4}$ with 102.4 ns mean latency (240 ns max) — well within the $1\,\mu ext{s}$ quantum cycle limit.

4. 100-Round Continuous Streaming Memory Survival: We validate continuous operation across $R=100$ measurement rounds ($100\,\mu ext{s}$ at $1\,\mu ext{s}$/cycle). Active temporal boundary residual propagation ($s_{ ext{res}}^{(k+1)} = s_{ ext{raw}}^{(k+1)} \oplus H_{ ext{boundary}} \hat{e}_{ ext{commit}}^{(k)}$) maintains 98.1% logical survival at 100 rounds ($F_{ ext{log}} = 0.9981$ at 10 µs, 0.9812 at 100 µs), whereas boundary-agnostic decoding collapses to 22.1% ($4.4 imes$ degradation).

5. Empirical IBM Quantum Heron r2 Calibration Benchmark: Evaluating our decoder against the empirical calibration snapshot of IBM's flagship 133-qubit Heron r2 processor ($p_{ ext{2Q}} = 2.1 imes 10^{-3}, \eta = 50.0, p_{ ext{RO}} = 8.0 imes 10^{-3}$) confirms convergence in only 3.85 mean iterations — the fastest among all 6 models tested (vs. 9.16 for Classical Min-Sum, 11.86 for Normalized Min-Sum, and 8.86 for BP+OSD-CS-10) with $ ext{DSP}=0$.

6. Hardware Synthesis & 10,000-Shot Co-Simulation Parity: We synthesize our complete architecture in SystemVerilog for AMD UltraScale+ XCKU040 FPGA ($f_{ ext{clk}} = 250 ext{ MHz}$, $WNS = +0.18 ext{ ns}$, 28,204 LUTs, 0 DSPs), confirming 10,000-shot bit-exact hardware-software co-simulation parity.

Scope Alignment with IEEE Transactions on Quantum Engineering:

This work addresses the intersection of quantum error-correcting codes, neural message-passing decoders, and real-time FPGA microarchitectures. It directly advances IEEE TQE's core mandate of bridging theoretical quantum information science and scalable physical quantum hardware engineering.

Editorial & Ethical Declarations:

1. **Originality:** This manuscript represents original research and is not currently published, accepted, or under concurrent evaluation in any other journal or conference.
2. **Software & Data Availability:** In accordance with IEEE open science standards, the complete implementation is publicly available as an open-source Python library (`orbit-tied-qnbp` v1.0.0 on PyPI and GitHub: `github.com/NazmulHasanNihal/orbit-tied-qnbp-sliding-window-bb`), including all synthesizeable SystemVerilog RTL, testbenches, and dataset generation scripts.
3. **Conflict of Interest:** The author certifies that there are no financial or commercial conflicts of interest.

Suggested Expert Reviewers:

Reviewer	Institution & Contact	Expertise & Rationale
Dr. Sergey Bravyi	IBM Quantum, T.J. Watson Research `sbravyi@us.ibm.com`	Pioneer of Bivariate-Bicycle QLDPC codes and gross codes.
Dr. Nicolas Delfosse	IonQ (formerly Microsoft Quantum) `ndelfosse@ionq.com`	Expert in sliding-window decoding, Union-Find, and LDPC Tanner graph symmetries.
Dr. Joschka Roffe	Freeform Labs / Univ. of Strathclyde `joschka@roffe.eu`	Developer of BP-OSD and expert in belief propagation on quantum codes.
Prof. Barbara Terhal	Delft University of Technology / QuTech `b.m.terhal@tudelft.nl`	Leader in fault-tolerant architectures and real-time QEC decoding limits.

Thank you for your time, consideration, and management of the peer-review process.

Sincerely,

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